GRADE 10 / MYP 5 • FORCE & MOTION

Toy-car investigation: measure motion

Pre-run to practise B/C assessment • Measure a car released from rest down a shallow incline.

TODAY'S GOALS

Collect and represent your own measurements

- Record elapsed time at marked positions.
- Repeat the run five times.
- Convert every position to metres and calculate mean times.
- Plot a smooth x–t graph manually.

INVESTIGATION QUESTION

Think about what we can measure

- How does the car’s position change with elapsed time?
- Use a fixed, shallow incline and the same toy car.
- Observe the car crossing distance marks along the ramp.

EQUIPMENT & ROLES

Prepare before the first run

EQUIPMENT

Toy car • shallow incline • paper scale • ruler • stopwatch • A4 graph paper

TEAM ROLES

Releaser • timekeepers / observers • data recorder.

Agree what counts as crossing a mark.

SET THE COORDINATE SYSTEM

A single marked straight path

- The release mark is x = 0.00 m.
- The positive direction is down the ramp.
- Mark positions every 5 cm (0.05 m) to 0.20 m.
- Record all times in seconds since release.

COLLECT FIVE TRIALS

Keep the raw observations

- Release the car from rest without a push.
- Record when the front of the car crosses each mark.
- Reset and repeat five times.
- Keep all measurements, including unusual readings.

EQUATIONS

Process each position

$$\text{mean time} = (t_1 + t_2 + t_3 + t_4 + t_5) / 5$$

Use the five times for the SAME position marker.

Record the mean elapsed time in seconds (s).

MANUAL GRAPH

Position against mean elapsed time

- Time, t (s) is on the horizontal axis.
- Position, x (m) is on the vertical axis.
- Plot points on A4 graph paper.
- Draw a smooth curve of best fit.

ESTIMATE VELOCITY

Gradient of the x–t curve

- Draw an early tangent to the smooth curve.
- Take two widely spaced points on the tangent.
- Calculate Δx / Δt in m/s.
- Repeat near the final measured time.

EXPLAIN WHAT YOU SEE

Compare two slopes

EARLY TANGENT

What does its slope tell you about the car’s velocity?

LATE TANGENT

How does this slope compare? What changed during the journey?

HAND IN / KEEP SAFE

You will need this tomorrow

- Raw results: all five trials.
- Mean-time table.
- Your manually plotted x–t graph with two tangents.
- Your calculations, with units.

LESSON SUMMARY

From motion to evidence

- Choose a clear origin and positive direction.
- Record repeated quantitative observations.
- Graph x against time in SI units.
- Use the graph gradient at various times to estimate velocity at those times.